Uncertainty Prediction Plots (1FA006)

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The Jupyter notebook can be found in this GitHub repository.

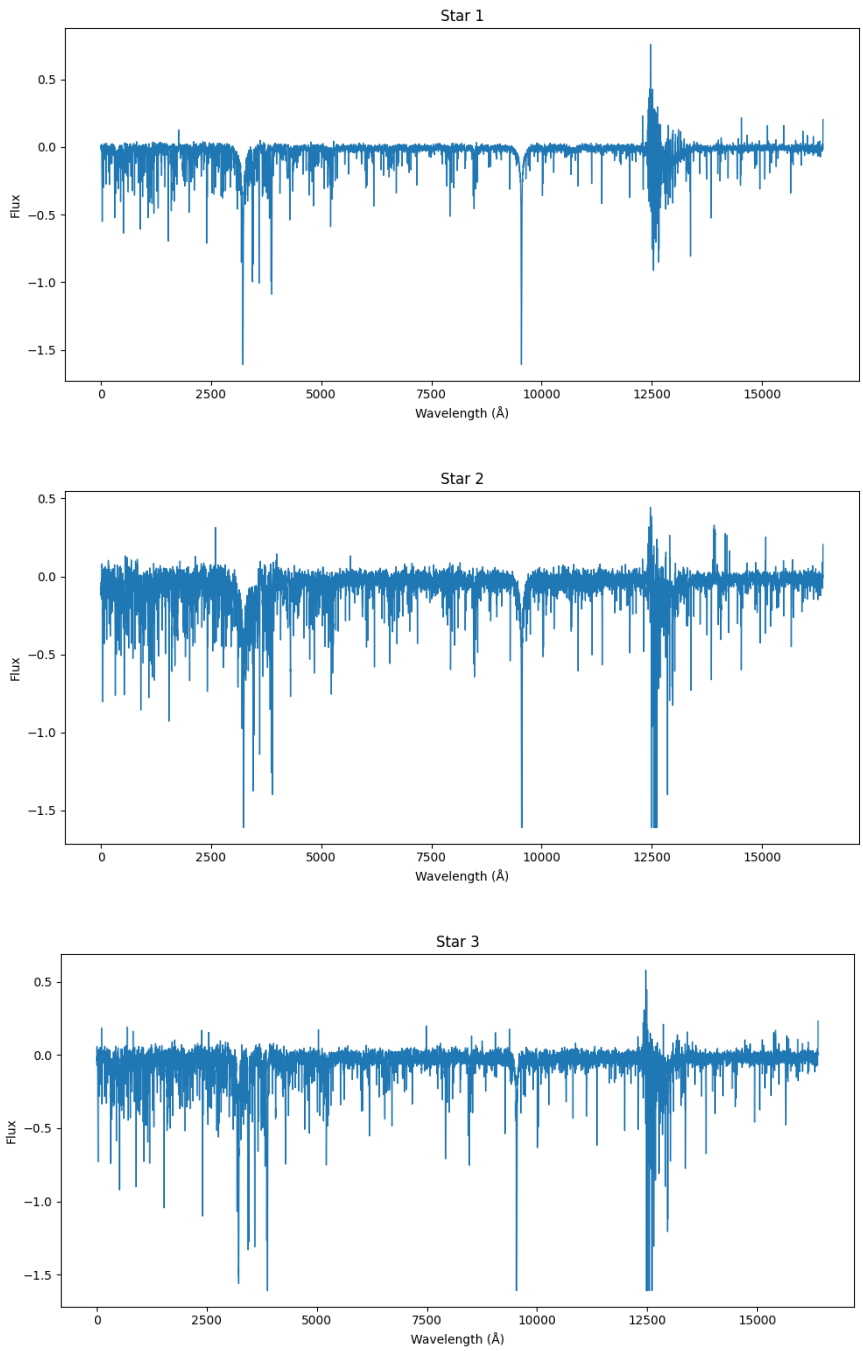

Figure 1: Examples of stellar spectra from the GALAH DR4 Dataset.

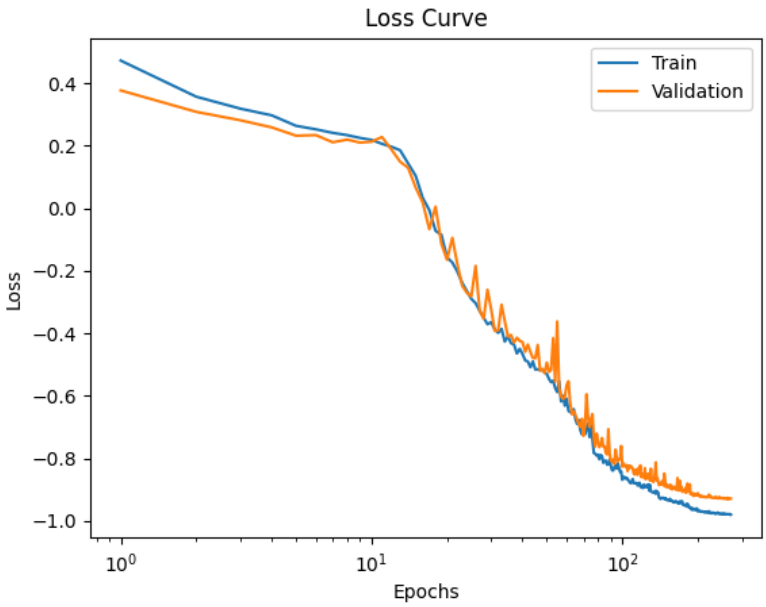

Figure 2: Training and Validation Loss v/s Epochs. The training and validation losses, although jittery, stabilize at around epoch 200 with an almost constant generalization error.

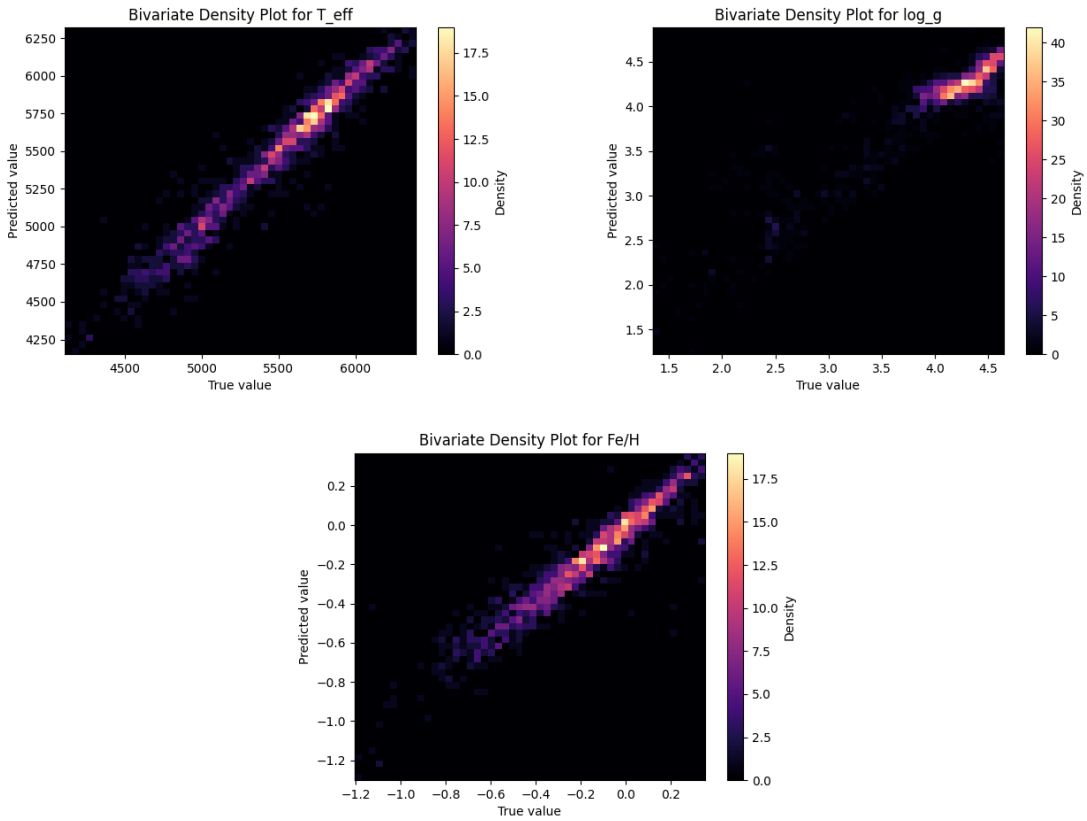

Figure 3: Bivariate histograms for each target label. This helps us visualize how "correct" the predictions of target values are. From these plots, we can tell that the model performance is good because a high density of data points lie around the y=x line. It is also interesting to see, in the $log(g)$ plot, many of the points are concentrated between 4.0 and 4.5 but still roughly follow the y=x trend.

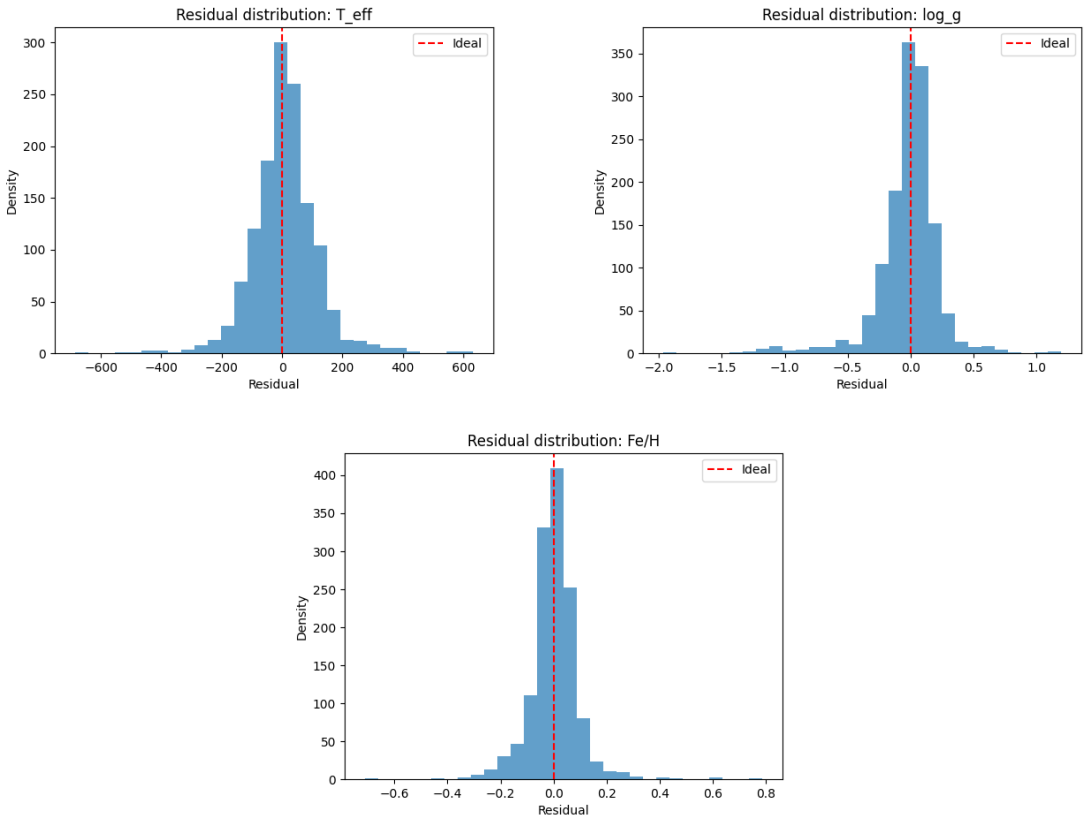

Figure 4: Residual histograms of each target label. This helps us visualize how the residuals are spread out. From these plots, we can tell that the residuals are centered around zero which means most of the points have a low residual error. This shows that the model is indeed good.

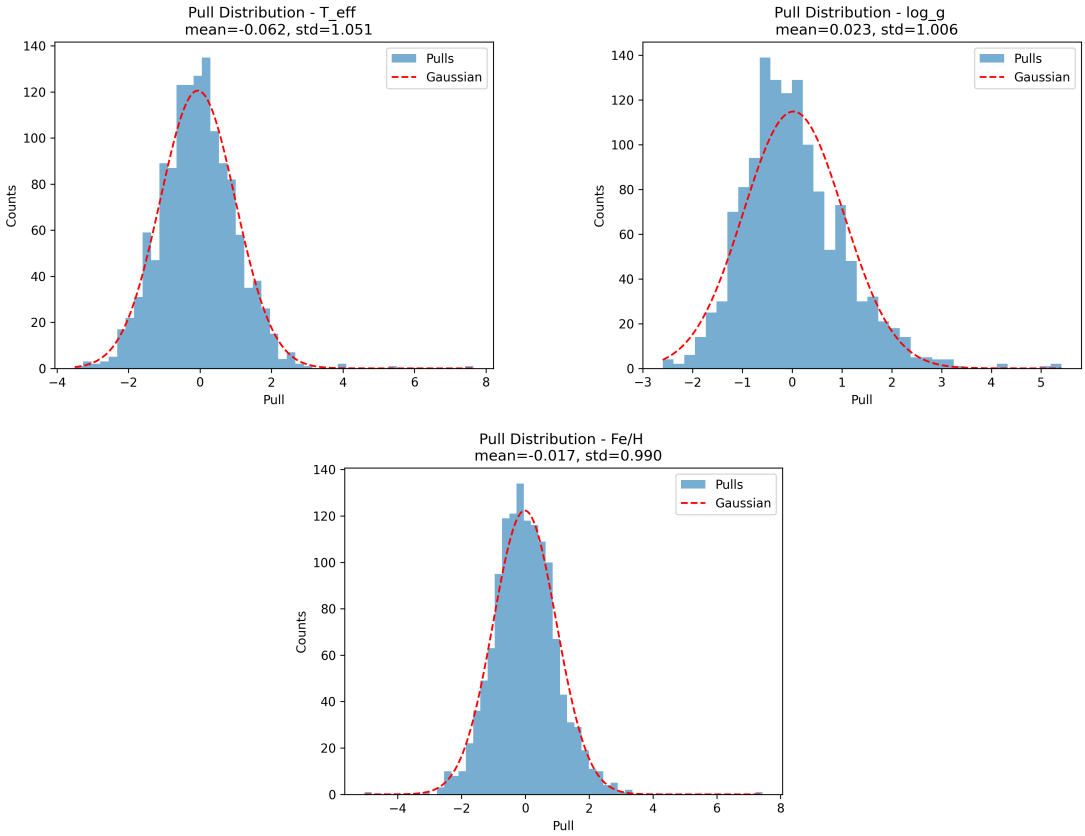

Figure 5: Pull distributions for each target label. This helps us visualize how "correct" the predictions of uncertainties are. From the plots above, we can see that the distributions are nearly gaussian with means and standard deviations around the ideal values.